The Scope of a Pass

An executable laboratory for warranted inference in AI research workflows

1. What travels with a checked result?

An AI analyst can calculate the right number, a checker can correctly approve the computation, and a later interpreter can still lack the evidence needed to use that number scientifically. A training–productivity difference, for example, supports a causal claim only under an appropriate design and assumptions. Successful arithmetic does not supply those premises. When research is divided among AI components, which information must accompany a checked result, and when should its recipient retrieve missing evidence before drawing a conclusion?

We propose Guard Lab, an executable laboratory for answering that question. It generates organizational research tasks with exact answers and inspectable limits on inference, then runs AI analysis, numerical checking, interpretation and evidence retrieval. Its central measurement distinguishes two outcomes: whether a conclusion is justified by the information its author received, and whether the research system completes the question its full source evidence can answer. A system can be appropriately cautious yet leave an answerable task unfinished; it can also repeat the correct source value without possessing its justification.

This is a problem of organizational control: a principal evaluates delegated work through records and checks that reveal only part of its quality (Ouchi 1979). In AI research production, the information connecting components can be designed and experimentally replayed. The intended contribution is a reusable instrument for studying how computation, verification and scientific justification remain connected across those boundaries.

The division of responsibility is consequential. A numerical checker can fulfill its assigned task while leaving the interpreter responsible for establishing identification. A project lead can preserve that responsibility through complete records, selective transfers or a route for requesting missing evidence. Guard Lab makes those arrangements experimental objects. AI enables repeated execution of the same research commission under different information policies, with observable intermediate artifacts and a separately checkable scientific outcome.

2. Answers and justified non-answers

The current generator implements four mathematical task structures: descriptive risk differences, causal effects under explicit random assignment, causal effects under explicit conditional exchangeability, and causal questions without supplied identifying restrictions. Constructed settings include training, retention, mentoring and process adoption. Domains rotate across structures. Integer weights encode exact population probability distributions, so the initial instrument separates identification and arithmetic from finite-sample uncertainty.

A rational-arithmetic oracle calculates each reference answer from the requested quantity and the supplied assumptions. For a nonidentified causal question, it constructs two potential-outcome populations with identical observed probabilities and different intervention effects. A separate implementation enumerates those populations, checks the observed laws and interventions, and verifies the actual answer payload. A justified non-answer is therefore supported by an explicit counterexample to uniqueness. These are established identification results (Hernán and Robins 2020), used as experimental controls.

The answer key is indexed to the recipient's information. Retaining an identifying record can support a causal point answer; withholding it can leave that answer unidentified even though the physical population is unchanged. Retrieving the archived record can restore the premise. A descriptive question remains answerable without causal identification, while a record explicitly supplying no identifying restriction cannot manufacture a causal answer. Correct nonidentification completes a question when its full source lacks identifying restrictions; it leaves the source task incomplete when available archived evidence would identify the requested effect. The grader evaluates constrained claim types and values; no model judges its own scientific success.

3. An implemented research workflow and its comparison

We built and ran a locally frozen development experiment using GPT-6 Astra and GPT-6.1 Sol, both at high reasoning effort through the same Codex client. Twelve newly parameterized tasks span the four structures. Each model produces twelve analysis programs and selects supporting records for a compact handoff. The harness executes each program in a restricted arithmetic process. A numerical checker reviews the artifact and may supply a replacement program. The resulting receipt names the artifact and the checks actually performed; it does not certify identification.

Six policies then expose the resulting artifact to an interpreter. A single-researcher baseline retains the complete source and a cumulative transcript of analysis and self-checking. A delegated baseline passes complete source records between fresh role contexts. A compact delegated policy passes the records the analyst selected, allowing an omission to arise from its own choice. Three separate stress policies deliberately remove the design record and present either a general pass message, a pass message naming its numerical limits, or general approval with the same inferential caution presented separately.

Each interpreter can finalize a constrained scientific answer or request the archived design record. The harness returns that record verbatim and permits one final decision. This is an executed analysis–checking–interpretation workflow with a bounded information-gathering action. It does not depend on hand-written model answers. The natural-selection policy and intentional-removal policies remain separate, so an engineered loss is never presented as a spontaneous failure.

The design contains 144 receiver episodes, built from 24 model-generated source analyses. Upstream artifacts are shared within model–task blocks for matched comparisons. Both architectures receive the same original task, model and reasoning setting, with the same stage opportunities. The single-researcher baseline uses explicit transcript replay across fresh CLI calls. Prompt length and realized reasoning tokens are not equalized; architecture comparisons concern these information policies, rather than a pure effect of agent count. Every planned episode, repair, missing response and tool violation remains in the accounting.

4. What the workflow experiment establishes

The completed experiment retained all 144 receiver episodes and 270 logical model calls. All 144 final claim types and values were warranted by received evidence. 24/24 initial programs executed successfully; 48/48 reviewed artifacts passed the arithmetic check. Analysts included the design record in 24/24 compact selections. Astra completed 12/12 source tasks in the shared-transcript baseline, 12/12 with complete delegated records, and 12/12 with analyst-selected records; Sol completed 12/12 source tasks in the shared-transcript baseline, 12/12 with complete delegated records, and 12/12 with analyst-selected records.

Astra: general 12/12 completed, 9 retrievals, scoped 12/12 completed, 9 retrievals, separate reminder 12/12 completed, 9 retrievals; Sol: general 12/12 completed, 9 retrievals, scoped 12/12 completed, 9 retrievals, separate reminder 12/12 completed, 9 retrievals.

Across the 72 intentional-removal episodes, interpreters requested the archive 54 times; final decisions included 0 unsupported causal claims and 0 scientifically supported but incomplete answers. These counts describe the fixed task set, with models reported separately in the supplement. The three stress-test messages produced identical completion, support, retrieval and unsupported-claim counts within each model. These cases supply no evidence of an advantage from attaching the caution to a verification receipt.

The frozen joint contract accepted 134/144 responses: its literal record-ID check flagged ten citations to the received population_table field. A disclosed post-run audit preserves those flags and separately reports 144/144 valid scientific answers; none cited unavailable data. The supplement retains all prior pilots and new trajectories. Offline reproduction recomputes reference checks and regrades saved responses; new model generation requires separate access.

The workflow result redirects the motivating argument. In this setting, capable systems preserved the design record or recovered it before committing. Retained-record policies required no archive requests; deliberate removal produced requests on all 54 causal-question episodes, including those whose source supplied no identifying restriction. Numerical approval did not substitute for a missing premise. This provides a positive control: the instrument recognizes successful evidence gathering as well as correct conclusions. An evaluator scoring only caution would miss whether an answerable question was completed; an evaluator scoring only the final number would miss whether its justification was ever available.

5. The contribution and its nearest alternatives

Generated causal-reasoning benchmarks already provide formal questions and answers (Jin et al. 2023). Scientific release gates and research-error benchmarks already evaluate whether workflows should release their outputs (Acharya et al. 2026; Willner and Yanagizawa-Drott 2026). Multi-agent failure taxonomies, strong single-agent comparisons and cross-domain scientific workflows establish that coordination and information integration have consequential limits (Cemri et al. 2025; Kim et al. 2026; Wong and Buehler 2026). We do not claim novelty for discovering communication problems or comparing agent architectures.

ClaimArena-LLM is especially close: it tracks unsupported causal propositions, evidence references and evaluator contamination in structured adversarial debates (Anghel et al. 2026). Guard Lab asks a different question about research production: how a numerically correct artifact becomes, remains or ceases to be sufficient for the requested inference as available methodological evidence changes. Its reference answers include quantitatively identified effects and nonidentification witnessed by observationally equivalent populations. Its workflow additionally distinguishes justified refusal from supported task completion and records whether missing justification is recovered.

The proposed advance is the ability to manipulate and verify those scientific boundaries within executed research workflows. That capability must earn its value through informative comparisons. The present experiment supplies an auditable starting point, including preserved successes and null contrasts. A blanket claim that verification creates false assurance would exceed this evidence.

6. Development, risks and adoption

The largest remaining risk is transfer. These tasks use short, explicitly labeled source records, four identification structures and two models from one provider. Their correct behavior does not establish reliability in unrestricted research. Mathematical checks were authored within the same development process and do not replace independent methodological review. The record-selection policy is structured rather than free-prose summarization; intentional deletion is a stress intervention. The single-researcher comparison is a transcript-retaining control, not an unrestricted autonomous scientist.

January development will freeze independently reviewed extensions to at least eight mathematical structures, including sample-versus-target-population questions where causal identification is not the issue. February will deliver a broader workflow demonstration for workshop review. March and April will test longer records, less explicit assumptions, alternative providers and more demanding evidence-retrieval costs. Task structures will be reserved before tuning, with seeds and paraphrases grouped under their originating argument. Finite-sample extensions will require an explicit uncertainty target rather than reuse the population oracle unchanged.

The next comparisons will vary who owns the responsibility for retrieving scientific premises and how that responsibility survives several handoffs. Evidence can be retained in full, summarized through a defined interface, or available only through a costly lookup. Holding the source question and permissible answer fixed will distinguish a control that preserves useful inference from one that merely encourages refusal. If stronger systems remain successful, the release will report the tested boundary and retrieval burden, without manufacturing a failure claim.

May will release a documented task specification, generator, evaluation interface, executable workflows and complete trajectories with evidence-specific conclusions. All current data are synthetic; saved results can be reproduced without private corpora, a GPU or a model account. Research teams could use the instrument to evaluate a proposed checking or handoff policy before adopting it, examining whether it preserves useful conclusions, prompts appropriate evidence requests and avoids unwarranted claims. Success would make the support for a scientific conclusion an inspectable property of AI research production, rather than an inference drawn from a successful computational check.

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